Instrument scale: levels and natural logarithms

Control: Historical instrument scale — Levels or Natural log.

1. Scale is applied after the historical instrument is constructed. The matching method first produces a historical population total or a historical density. Let $P_{g,t}$ denote the resulting population in people, and $v_{g,t}$ a resulting density in people per square kilometre. The instrument $z_{g,t}$ is

	Levels	Natural log
Population	$P_{g,t}/1{,}000$	$\log P_{g,t}$
Density	$v_{g,t}$	$\log v_{g,t}$.

The population total is used by county matching and area weighting. Maximum-density overlap uses historical density under the Employment density approach, and the full population of the selected county under Employment.

2. A logarithm changes the relative importance of large values. With a level instrument, increasing population from 10,000 to 20,000 changes the instrument by 10, while increasing it from 100,000 to 200,000 changes it by 100. With a log instrument, both changes equal $\log 2$. Thus, a log gives equal instrument changes to equal proportional population changes. It can change which observations provide most of the useful variation after controls and fixed effects, and can change the IV estimate.

Dividing population by 1,000 is different: it only changes the unit of measurement. In an otherwise unchanged linear IV regression, this rescaling does not change the estimated density effect, fitted values or instrument-strength statistic. It changes the reported first-stage coefficient's units.

3. Zero observations require attention. The Levels transformation preserves a recorded zero. Other eligibility rules still apply: the original overlap methods use positive historical densities. Natural log requires a positive value, so zero or negative values are excluded; the app reports these exclusions. It does not add one before taking the logarithm. Missing values remain missing under both options. Consequently, a levels–log comparison can change both the instrument and the estimation sample.

4. The modern regression and historical CH option are separate. Choosing a level instrument does not remove the logarithm from modern employment or average employment density on the right-hand side. The scale control concerns the historical instrument. The State-level Historical CH density index is already a population-weighted average of historical log densities. It enters directly and is unchanged by this selector. County-adjusted CH also uses population in levels for its linked county calibration, separately from the regional instrument-scale choice.